

PROJECT REPORT

ADVANCED BMI CALCULATOR WITH DATA ANALYTICS

Submitted By

Siddhnath Prajapati

Abstract

The Advanced BMI Calculator is a desktop-based healthcare application developed using Python. The application calculates the Body Mass Index (BMI) of users based on their weight and height. It further categorizes users into different health classifications such as Underweight, Normal Weight, Overweight, and Obese.

The system stores user information in a database, allowing historical record maintenance and retrieval. It also provides graphical trend analysis and statistical insights to help users monitor their health status over time.

Chapter 1: Introduction

Body Mass Index (BMI) is a widely used measure to assess an individual's body weight relative to height. It helps determine whether a person is underweight, normal weight, overweight, or obese.

This project automates the BMI calculation process and provides advanced functionalities such as data storage, historical tracking, and graphical analysis.

Chapter 2: Objectives

The primary objectives of the project are:

- To calculate BMI accurately.
- To provide a user-friendly graphical interface.
- To maintain historical BMI records.
- To visualize BMI trends through graphs.
- To generate statistical analysis of BMI data.
- To improve healthcare awareness using technology.

Chapter 3: Software Requirements

Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB Minimum
- Storage: 100 MB Free Space

Software Requirements

- Python 3.x
 - Tkinter
 - SQLite3
 - Pandas
 - Matplotlib
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Chapter 4: System Design

Input

- User Name
- Weight (kg)
- Height (m)

Processing

BMI = Weight / Height²

BMI Classification:

- Underweight (<18.5)
- Normal Weight (18.5–24.9)
- Overweight (25–29.9)
- Obese (≥ 30)

Output

- BMI Value
 - Health Category
 - Stored Database Record
 - Statistical Analysis
 - Trend Graph
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Chapter 5: Methodology

The application follows the following process:

1. User enters personal details.
 2. BMI is calculated.
 3. BMI category is determined.
 4. Data is stored in SQLite database.
 5. Historical records are maintained.
 6. Statistical analysis is generated.
 7. Graphical visualization is displayed.
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Chapter 6: Modules Description

BMI Calculation Module

Calculates BMI using standard formula.

Database Module

Stores user records permanently.

Search Module

Searches historical records.

Statistics Module

Computes:

- Average BMI
- Highest BMI
- Lowest BMI

Graph Module

Generates trend analysis charts using Matplotlib.

Chapter 7: Results

The application successfully:

- Calculates BMI accurately.
- Stores multiple user records.
- Retrieves historical information.
- Displays statistical analysis.
- Generates BMI trend graphs.

The developed system provides an efficient and reliable solution for BMI monitoring and health tracking.

Chapter 8: Advantages

- Easy to Use
 - Accurate BMI Calculation
 - Permanent Data Storage
 - Health Trend Monitoring
 - Interactive GUI
 - Real-Time Statistics
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Chapter 9: Future Scope

Future enhancements may include:

- Cloud-Based Storage
 - Mobile Application Version
 - Health Recommendation System
 - AI-Based Health Prediction
 - User Authentication System
 - PDF Report Generation
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Chapter 10: Conclusion

The Advanced BMI Calculator successfully combines healthcare monitoring with modern software technologies. The integration of GUI design, database management, statistical analysis, and graphical visualization makes the application more effective than a traditional BMI calculator.

The project demonstrates practical implementation of Python programming concepts and provides valuable experience in desktop application development.

References

1. Python Official Documentation
 2. Tkinter Documentation
 3. SQLite Documentation
 4. Pandas Documentation
 5. Matplotlib Documentation
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Author

Siddhnath Prajapati

Python Developer | Software Engineering Student